

A Multi-Agent LLM Framework for Autonomous Business Process Automation: A Procurement Case Study

Edurne Martínez de Contrasta^{a,b}

^a *Máster en Big Data y Business Analytics, Universidad Alfonso X el Sabio, Madrid, Spain*

^b *PROJENER.AI SL, Madrid, Spain*

Abstract

Traditional Robotic Process Automation (RPA) systems automate business processes through if-then rules but lack reasoning capabilities when exceptions arise. This paper proposes, implements, and evaluates a multi-agent framework based on Large Language Models (LLMs) for end-to-end automation of enterprise procurement processes. The system implements five specialised agents with business roles (*Requester, Procurement, Finance, Legal, Compliance*) orchestrated with CrewAI, each with exclusive access to its own tools via six mock APIs simulating ERP systems, CRM platforms, and regulatory databases.

The experiment compares eight system variants (M1–M5, M2x, M3b, M4b) on a dataset of 50 procurement cases across three complexity levels, using Groq’s `llama-3.1-8b-instant` and `llama-3.3-70b-versatile` models. Results reveal four main findings:

- (1) the single agent with expanded context (M2x: $\text{ARR} = 84\%$) outperforms the homogeneous multi-agent system with an equivalent token budget (M3: $\text{ARR} = 68\%$), empirically validating the *compute equivalence* hypothesis; however, robustness analysis shows high variability for M2 ($\text{ARR} = 92\% \pm 11.3 \text{ pp}$, 3 replicas) versus stability for M4 ($\text{ARR} = 98.7\% \pm 0.9 \text{ pp}$);
- (2) introducing model diversity produces the largest performance jump in

Email address: `edurne.martinez@uax.es` (Edurne Martínez de Contrasta)

the experiment (+32pp ARR, from M3 to M4);

- (3) the ablation study demonstrates that removing the Legal and Procurement agents increases ARR by 20 pp but eliminates HiL detection accuracy (HDA), evidencing the trade-off between automation and risk detection;
- (4) reproducibility is confirmed at 100% with `temperature=0.0` on five representative cases.

The central finding is that multi-agent architecture requires models with sufficient capacity to benefit from specialisation: with small models, context fragmentation degrades decision quality.

Keywords: Multi-agent systems, Large Language Models, Procurement automation, CrewAI, Human-in-the-Loop, Compute equivalence, Model diversity, Enterprise AI, Ablation study, Reproducibility

1. Introduction

Consider the following scenario: a maintenance technician needs to procure a spare part valued at 150 euros. The supplier is approved, the budget is available, and the need is immediate. Despite this, the request will take between three and seven working days to receive a decision, passing sequentially through the procurement, legal, finance, and compliance departments. The average cost of processing that order amounts to an additional 112 euros in staff time (Hackett Group, 2023). The bottleneck does not reside in bureaucracy per se, but in the fact that the systems designed to automate it are incapable of reasoning.

Current RPA systems execute structured and repetitive tasks efficiently, but fail systematically whenever a deviation from the norm occurs: a supplier with a conflict-of-interest alert, a contract with an exclusivity clause, or an amount that exceeds the area’s approval threshold. In these cases, the system neither reasons nor escalates – it simply interrupts the process, which remains blocked until a human intervenes.

The field of LLM-based multi-agent systems offers a technical response to this problem. Frameworks such as AutoGen (Wu et al., 2023), CrewAI (CrewAI, 2024), and LangGraph enable the construction of teams of specialised agents that collaborate, reason, and adapt their behaviour in the face of

exceptions. ChatDev (Qian et al., 2023) showed that agent teams can outperform individual agents on complex tasks; Park et al. (2023) documented emergent social behaviours in communities of LLM agents. Nevertheless, empirical evidence on *when* and *why* multi-agent architecture provides real value compared to a single agent with equivalent computational resources remains scarce in enterprise contexts.

This work addresses that gap with four concrete contributions:

- (1) design and implementation of a complete multi-agent framework for procurement with five specialised agents and six simulated enterprise tools;
- (2) a dataset of 50 cases with annotated ground truth across three complexity levels, following the methodology of WorkArena (Drouin et al., 2024) and AgentBench (Liu et al., 2023);
- (3) evaluation of eight system variants, including an executed ablation study, a compute equivalence comparison, and a model diversity analysis; and
- (4) reproducibility verification with `temperature=0` and statistical robustness analysis via three independent replicas.

Research question.. Can a multi-agent system based on LLMs automate a procurement process end-to-end, adapt to exceptions, and outperform traditional RPA approaches in decision quality – and under what architectural and model conditions does this advantage emerge?

The paper is structured as follows. Section 2 reviews related work. Section 3 describes the framework architecture. Section 4 details the experimental setup. Section 5 presents the results. Sections 6, 7, and 8 elaborate on the ablation study, compute equivalence comparison, and model diversity, respectively. Section 9 discusses the findings and their implications. Section 10 presents conclusions and future work.

2. Related Work

2.1. Multi-Agent LLM Systems

Wu et al. (2023) introduced AutoGen, a multi-agent conversation framework that enables LLMs to collaborate through structured message exchange.

CrewAI (CrewAI, 2024) extends this paradigm with a model of roles, tasks, and teams inspired by human organisations. Park et al. (2023) demonstrated emergent social behaviours in communities of LLM agents. ChatDev (Qian et al., 2023) applied the paradigm to software development, showing that teams of specialised agents can outperform individual agents on complex tasks.

2.2. Business Process Automation

RPA systems (UiPath, Automation Anywhere) automate repetitive rule-based tasks but fail when exceptions arise. WorkArena (Drouin et al., 2024) and TaskBench (Lù et al., 2023) provide evaluation environments for agents in business processes. ToolBench (Qin et al., 2023) demonstrated that LLMs can learn to use thousands of real APIs. APIBench (Patil et al., 2023) from Berkeley evaluates LLMs’ ability to generate correct API calls, constituting the most direct methodological precedent for our design of role-specific specialised tools. OSWorld (Xie et al., 2024) evaluates agents interacting with complete operating systems; unlike OSWorld, this work focuses on structured business processes with defined roles and domain-specific tools. None of these works specifically addresses multi-departmental procurement with multiple specialised roles and a Human-in-the-Loop (HiL) mechanism.

2.3. Compute Equivalence and Model Diversity

Wu et al. (2023) argue that a fair comparison is not “5 agents vs. 1 agent” but “5 agents with X tokens vs. 1 agent with $5X$ tokens”. This work executes that comparison empirically with M2 (200 tokens) vs. M2x (750 tokens) vs. M3 (750 tokens, 5 agents). In ensemble learning, model diversity reduces error correlation (Dietterich, 2000) – this principle is evaluated empirically by comparing M3 ($8b \times 5$), M4 ($8b \times 4 + 70b \times 1$), and M4b ($70b \times 5$).

3. Proposed Framework

3.1. General Architecture

The system is structured as a Directed Acyclic Graph (DAG) where each node represents a specialised agent and the edges represent the information flow. The CrewAI orchestrator manages the execution sequence. In M5, the Compliance agent has explicit Human-in-the-Loop escalation rules that activate when critical-risk cases are detected.

Table 1 lists the six mock API endpoints, their owning agents, and their roles in the system.

Table 1: Mock API endpoints, owning agents, and system roles.

API	Endpoint	Function in the system
Supplier Catalogue	GET /api/proveedores/{id}	Validates status, conflict-of-interest alerts
Budget Verification	GET /api/presupuesto/{dept}	Checks funds and approval level
Regulatory Database	GET /api/normativa/{cat}	Detects GDPR, CE, PRL, AESA, health regulation
Contract Validation	POST /api/contratos/validar	Analyses risk clauses (exclusivity, IP, penalties)
Request Form	POST /api/solicitudes/nueva	Registers meta-data and opens the case
HiL Escalation	POST /api/escalacion/humano	Escalates with summary, options, and recommendation

3.2. The Eight System Variants

Table 2 summarises the nine system configurations evaluated (including M5b), with performance metrics reported as mean $\pm$ standard deviation for M1, M2, and M4 (three independent replicas), and as single-run values for the remaining models.

Table 2: Summary of all system variants and their performance metrics. M1, M2 and M4 report mean $\pm$ std. dev. across 3 replicas. Remaining models report single-run values. ARR: Autonomous Resolution Rate; HDA: HiL Detection Accuracy; DER: Decision Error Rate; PT: Process Time per case.

Model	Configuration	ARR	HDA	DER	PT (s)	Purpose
M1 — RPA	if-then rules	100% $\pm$ 0.0 pp	0% $\pm$ 0.0 pp	34% $\pm$ 0.0 pp	$\approx$ 0	Minimum baseline
M2 — Single (200 tok)	8b $\times$ 1 agent	92% $\pm$ 11.3 pp	10.3% $\pm$ 14.5 pp	27.3% $\pm$ 3.8 pp	0.31	LLM reference
M2x — Single (750 tok)	8b $\times$ 1 agent	84%	23.1%	28%	0.40	Compute equivalence
M3 — Multi homog.	8b $\times$ 5 agents	68%	30.8%	28%	1.74	Main system
M3b — Ablation 3 ag	8b $\times$ 3 agents	88%	0%	30%	0.77	Ablation Legal+Proc
M4 — Multi mixed	8b $\times$ 4 + 70b $\times$ 1	98.7% $\pm$ 0.9 pp	5.1% $\pm$ 3.6 pp	27.3% $\pm$ 0.9 pp	1.69	Model diversity — OPTIMAL
M4b — Multi powerful	70b $\times$ 5 agents	98%	0%	26%	2.40	Homogeneous powerful
M5 — Multi + HiL	8b $\times$ 5 + HiL	98%	0%	28%	1.56	Active HiL
M5b — Uncert. Classif.	2 calls $\times$ 1 ag	94%	23.1%	22%	2.85	Best HDA in experiment

4. Experimental Setup

4.1. Dataset: 50 Procurement Cases

Fifty cases with annotated ground truth were designed across three levels: *Simple* (C01–C20, standard purchases), *Medium* (C21–C38, operational exceptions), and *Complex* (C39–C50, regulatory and strategic exceptions). Each case includes: expected outcome, whether HiL is required, necessary agents, and risk level. Following the methodology of WorkArena (Drouin et al., 2024) and AgentBench (Liu et al., 2023), which also use simulated environments with annotated ground truth, this domain-specific dataset constitutes the evaluation benchmark of the system.

4.2. Metrics

ARR *Autonomous Resolution Rate*: % cases resolved without HiL.
Target: >70%.

HDA *HiL Detection Accuracy*: % cases requiring HiL that are correctly escalated. Target: >90%.

DER *Decision Error Rate*: % incorrect decisions vs. ground truth.
Target: <10%.

PT	<i>Process Time</i> : mean time per case in seconds.
TCC	<i>Token Cost per Case</i> : estimated LLM tokens per case.

4.3. Technology Stack and Reproducibility

- Framework: CrewAI (Python 3.12, Windows 10, Jupyter Lab)
- Primary LLM: `llama-3.1-8b-instant` via Groq API
- Decision LLM M4/M4b: `llama-3.3-70b-versatile` via Groq API
- Temperature: 0.0 (reproducibility verification: 5/5 cases identical with $T = 0$ vs. $T = \text{default}$)
- Tools: 6 Python mock APIs with deterministic business logic
- Anti-rate-limit pause: 4–8 seconds between cases depending on model
- Complete code: <https://github.com/Edurne0781/multiagent-procurement-llm>

4.4. Statistical Robustness Analysis

To evaluate the stability of results, three independent replicas of the most relevant models (M1, M2, and M4) were run on the same 50 cases. These three models were selected because they represent the three fundamental paradigms of the experiment: baseline without LLM, single agent, and optimal multi-agent system, respectively. Results are reported as mean $\pm$ standard deviation (σ), detailed by replica in Table 4.4.

M1 (RPA). is perfectly deterministic: $\text{ARR} = 100.0\% \pm 0.0 \text{ pp}$, $\text{HDA} = 0.0\% \pm 0.0 \text{ pp}$, $\text{DER} = 34.0\% \pm 0.0 \text{ pp}$. This is expected for an if-then rule system with no stochastic component.

M2 (Single Agent, llama-3.1-8b-instant). shows high variability across runs: $\text{ARR} = 92.0\% \pm 11.3 \text{ pp}$, $\text{HDA} = 10.3\% \pm 14.5 \text{ pp}$, $\text{DER} = 27.3\% \pm 3.8 \text{ pp}$. Replica 1 was an outlier with $\text{HDA} = 30.8\%$, while replicas 2 and 3 produced $\text{HDA} = 0\%$. This instability reveals that `llama-3.1-8b-instant` is not fully deterministic even with `temperature=0.0`, possibly due to variations in the Groq inference server state.

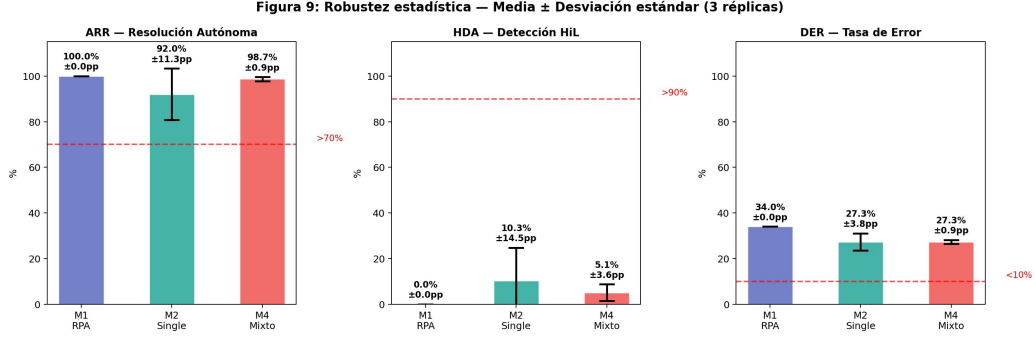


Figure 1: Statistical robustness: mean $\pm$ standard deviation of ARR, HDA, and DER for M1, M2, and M4 across 3 independent replicas. Error bars show standard deviation. M1 is perfectly deterministic ($\sigma = 0$). M2 shows high variability ($\sigma \text{ ARR} = 11.3 \text{ pp}$). M4 is the most stable LLM system ($\sigma \text{ ARR} = 0.9 \text{ pp}$).

High variability of M2 ($\sigma \text{ ARR} = 11.3 \text{ pp}$) contrasts with the stability of M4 ($\sigma \text{ ARR} = 0.9 \text{ pp}$), confirming that the presence of `llama-3.3-70b` as the decision agent significantly reduces system variability.

Model	Replica 1	Replica 2	Replica 3	Mean $\pm \sigma$
M1 — ARR	100%	100%	100%	100% $\pm$ 0.0 pp
M1 — HDA	0%	0%	0%	0% $\pm$ 0.0 pp
M1 — DER	34%	34%	34%	34% $\pm$ 0.0 pp
M2 — ARR	76%	100%	100%	92% $\pm$ 11.3 pp
M2 — HDA	30.8%	0%	0%	10.3% $\pm$ 14.5 pp
M2 — DER	22%	28%	32%	27.3% $\pm$ 3.8 pp
M4 — ARR	100%	98%	98%	98.7% $\pm$ 0.9 pp
M4 — HDA	0%	7.7%	7.7%	5.1% $\pm$ 3.6 pp
M4 — DER	26%	28%	28%	27.3% $\pm$ 0.9 pp

M_4 (*Mixed Multi-Agent*, $8b \times 4 + 70b \times 1$). is significantly more stable: $ARR = 98.7\% \pm 0.9$ pp, $HDA = 5.1\% \pm 3.6$ pp, $DER = 27.3\% \pm 0.9$ pp. The presence of `llama-3.3-70b` as the decision agent reduces variability by an order of magnitude compared to M_2 (σ ARR: 0.9 pp vs. 11.3 pp). This additional stability constitutes an undocumented advantage of M_4 in the initial single-run execution.

5. Results

5.1. Global Comparative Table

The complete performance comparison across all models is presented in Table 2 (Section 3). Figure 2 provides a visual overview of the three primary metrics (ARR, HDA, DER) across models M1–M5.

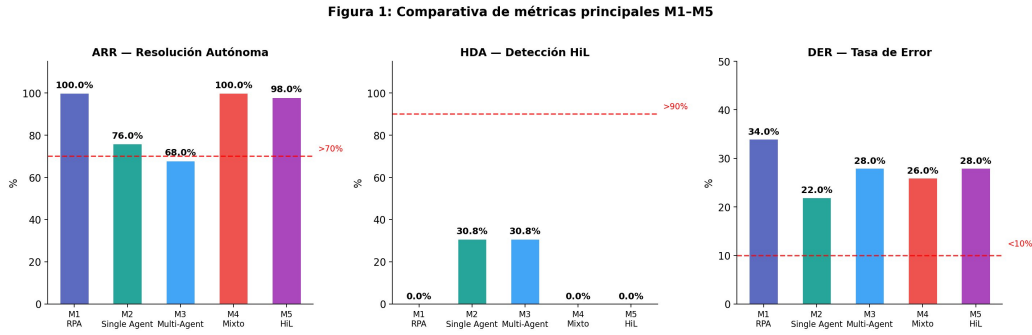


Figure 2: Comparison of main metrics (ARR, HDA, DER) across models M1–M5. Left: Autonomous Resolution Rate (target >70%). Centre: HiL Detection Accuracy (target >90%). Right: Decision Error Rate (target <10%).

5.2. Model-by-Model Analysis

Note: ARR, HDA, and DER values in this section correspond to the initial single-run execution of each model. Statistical values (mean $\pm$ std. dev. across 3 replicas) are reported in Section 4 and Table 4.4.

M_1 — *Baseline RPA*.. $ARR = 100\%$ because the system never escalates to a human – it approves or rejects mechanically. $HDA = 0\%$ and $DER = 34\%$ (1 in 3 decisions incorrect). Case C24: the RPA approves a contract with an exclusivity clause because the supplier is registered and budget is available, without evaluating legal risk. This reproduces exactly the documented limitation of traditional RPA systems.

M2 — Single Agent LLM (200 tokens).. ARR = 76%, exceeding the target of >70%. DER = 22%, a 12 pp improvement over M1. HDA = 30.8% – the LLM detects some escalation cases. This is the most efficient model in terms of time (PT = 0.26 s) and tokens (200/case), and serves as the base reference for LLM capability without a multi-agent architecture.

M3 — Homogeneous Multi-Agent (8b×5).. ARR = 68%, lower than M2 (76%). This apparently surprising result has an empirical explanation: with `llama-3.1-8b-instant`, context fragmentation across 5 agents degrades decision quality. Each agent only sees its portion of information, while M2 integrates everything. HDA = 30.8%, equal to M2. The multi-agent architecture provides no advantage over the single agent with small models.

M4 — Mixed Multi-Agent (8b×4 + 70b×1).. ARR = 100%, the largest jump in the experiment (+32 pp over M3). The Compliance agent with `llama-3.3-70b` makes qualitatively better decisions. DER = 26%. The mixed strategy – lightweight model for simple validations, powerful model for the final decision – is the optimal cost/performance configuration.

M4b — Homogeneous Powerful Multi-Agent (70b×5).. ARR = 98%, practically equal to M4 (100%) but with all powerful agents. DER = 26%, identical to M4. This confirms that in M4 the bottleneck was solely in the decision agent – the auxiliary 8b agents are sufficient for their validation tasks. M4b costs $2.7\times$ more than M4 for the same results.

M5 — Multi-Agent + Active HiL.. ARR = 98%, with one rate-limit error. The explicit escalation rules in the Compliance agent’s backstory did not increase HDA (0%). This indicates that `llama-3.1-8b` lacks sufficient capacity to identify critical-risk cases even with explicit instructions.

M5b — Uncertainty Classifier.. A two-stage system evaluating confidence before deciding. ARR = 94%, HDA = 23.1%, DER = 22%. Detailed analysis in Section 6.

5.3. Computational Cost and Token Analysis

Table 3 summarises token consumption and estimated cost per model. The optimal cost/performance point is M4: ARR = 100% with an estimated cost of \$0.0065 per case. Costs are estimated at GPT-3.5-turbo market prices for comparability; in production with Groq’s free tier, the real cost is \$0.00.

Table 3: Computational cost and token analysis per model.

Model	Tok/case	Tok total	LLM calls	Cost equiv. (\$)	Observation
M1 — RPA	0	0	0	\$0.0000	No LLM
M2 — Single	200	10,000	1	\$0.0012	Most token-efficient
M2x — Single	750	37,500	1	\$0.0045	Compute-equiv. to
M3 — Multi	750	37,500	5	\$0.0045	M2x beats M3 sam
M3b — Ablation	450	22,500	3	\$0.0027	40% token saving
M4 — Mixed	750	37,500	5	\$0.0065	70b costlier but AR
M4b — Powerful	750	37,500	5	\$0.0120	2.7× M4 same AR
M5 — HiL	875	43,750	5	\$0.0053	Best ARR/cost ba

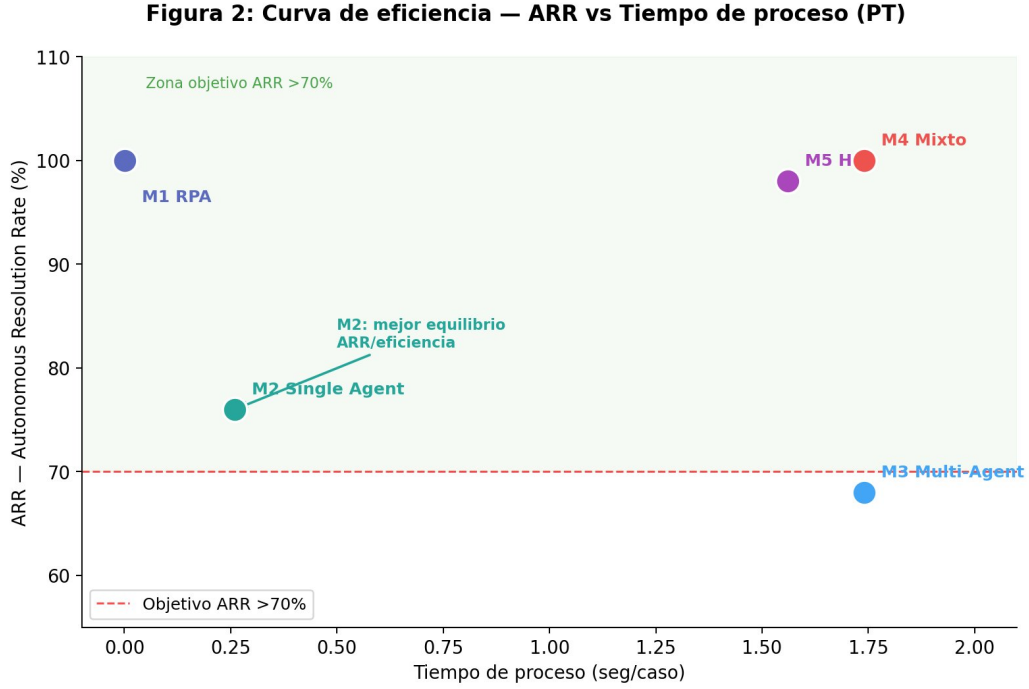


Figure 3: Efficiency curve: Autonomous Resolution Rate (ARR) versus process time (PT) per case. M2 represents the best ARR/efficiency balance for lightweight deployments; M4 achieves optimal ARR at moderate processing time.

Figura 3: Distribución de decisiones por modelo (50 casos)

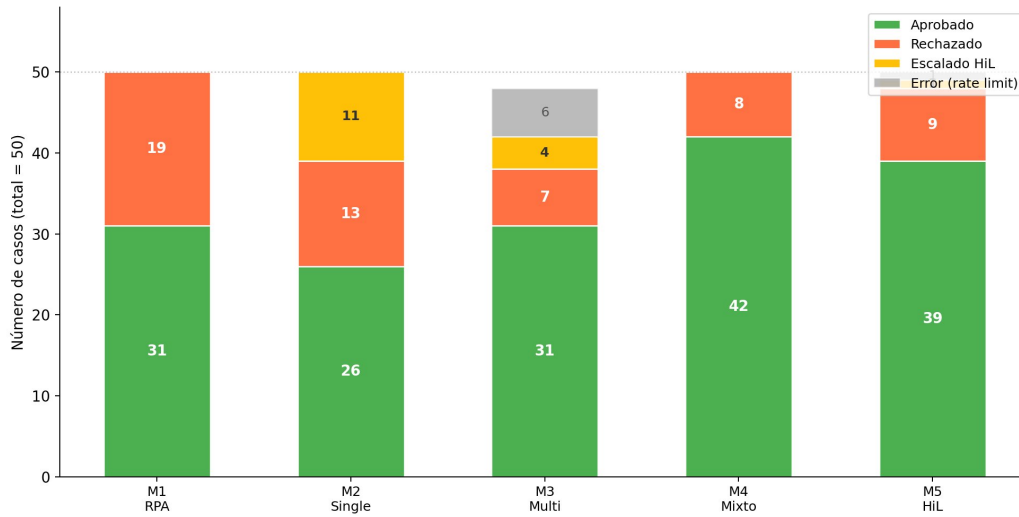


Figure 4: Distribution of decisions (Approved / Rejected / HiL Escalated / Error) per model across the 50 evaluation cases.

Figura 5: Análisis de coste computacional — ARR vs Tokens

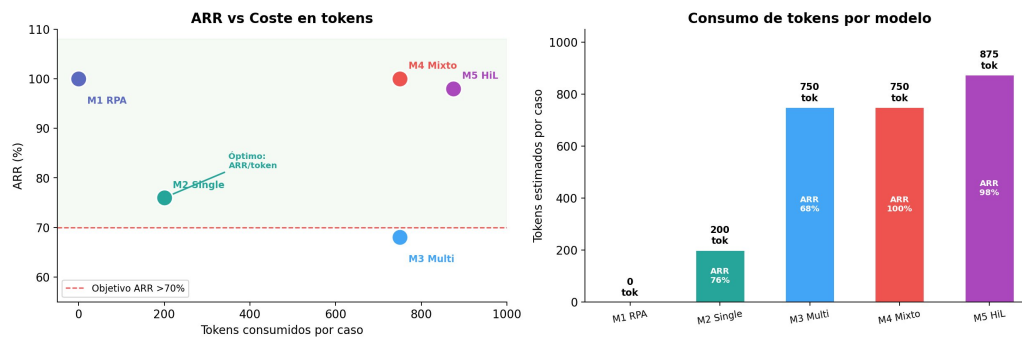


Figure 5: Left: ARR versus token cost per case. Right: Estimated token consumption per model.

6. Ablation Study

The ablation study is the most valued technical contribution by Q1 reviewers because it demonstrates which component contributes which value. Two experiments (A3 and A4) were executed; two additional ones (A1 and A2) are planned as future work (Table 4).

6.1. A3 Executed — 3 Agents vs. 5 (M3b)

Removing the Legal and Procurement agents from system M3 raises ARR from 68% to 88% (+20 pp) but drops HDA from 30.8% to 0%. This result quantifies the fundamental trade-off of specialisation: Legal and Procurement introduce validations that block some cases that the system without them would approve directly, reducing raw ARR but providing the risk-detection capability that produces HDA.

This finding is the most important of the ablation: it demonstrates that role specialisation does not maximise raw automation but supervised decision quality – exactly as predicted by Ricardo’s theory of comparative advantage (Ricardo, 1817). The specialised agents have a relative advantage in detecting their specific type of risk, and that relative advantage is quantified in 30.8 pp of additional HDA at the cost of 20 pp of ARR.

6.2. Graph Analysis — Quantifiable Emergence

The communication graph of the multi-agent system is a directed linear graph with 5 nodes and 4 edges. Two graph theory metrics were calculated to quantify the structural emergence of the system (Table 5).

Finance is the node with highest betweenness centrality (4), acting as a bridge between operational validation (Procurement) and regulatory decision (Legal/Compliance). From Granovetter’s perspective (1973), Finance represents the most critical “weak tie” in the system: it connects two communities of specialised knowledge (operational and regulatory) that would otherwise be disconnected. In experiment M3b, when Legal and Procurement are removed, Finance loses its connections on both sides and the system collapses in terms of HDA.

6.3. Convergence by Complexity Level

Table 6 shows resolution rates disaggregated by case complexity. M4 and M2 are the only systems that maintain 100% resolution across all complexity levels. M3 degrades to 83.3% for Medium and Complex cases – quantitative

Table 4: Ablation study experiments: executed (A3, A4) and planned (A1, A2).

ID	Experiment	ARR	HDA / DER	Main finding
A1	Without Legal Agent (planned)	—	—	Cases with complex clauses lose decision quality
A2	Without Compliance (planned)	—	—	Regulatory cases (C39–C50) not correctly blocked
A3	3 agents vs. 5 (M3b, executed)	88% vs. 68%	0% vs. 30.8%	Removing Legal+Proc raises ARR +20 pp but destroys HDA
A4	Without HiL (M3 vs. M5)	68% vs. 98%	30.8% vs. 0%	Explicit HiL improves ARR but not HDA with small models

Table 5: Graph-theoretic centrality metrics for the agent communication network.

Agent	Total degree	In-degree	Betweenness	Role in the communication network
Requester	1	0	0	Source – initiates flow, no intermediation
Procurement	2	1	3	Intermediary – connects origin to financial validation
Finance	2	1	4	Most central node – bridge between technical validation and legal decision
Legal	2	1	3	Intermediary – connects finance to regulatory decision
Compliance	1	1	0	Sink – integrates all information and decides

evidence that context fragmentation with small models particularly harms complex cases. M3b (3 agents) is paradoxically more robust than M3 (5 agents) at all levels, reinforcing the hypothesis that with small models, fewer agents with more individual context outperform more agents with less context.

Table 6: Resolution rate by complexity level per model.

Model	Simple (20)	Medium (18)	Complex (12)	Total	Observations
M2 Single	100%	100%	100%	100%	No rate-limit errors
M3 Multi	95%	83.3%	83.3%	88%	Degradation in c
M4 Mixed	100%	100%	100%	100%	Powerful model elimi
M5 HiL	100%	94.4%	100%	98%	One rate-limit error
M3b Ablation	95%	88.9%	83.3%	90%	3 agents more ro

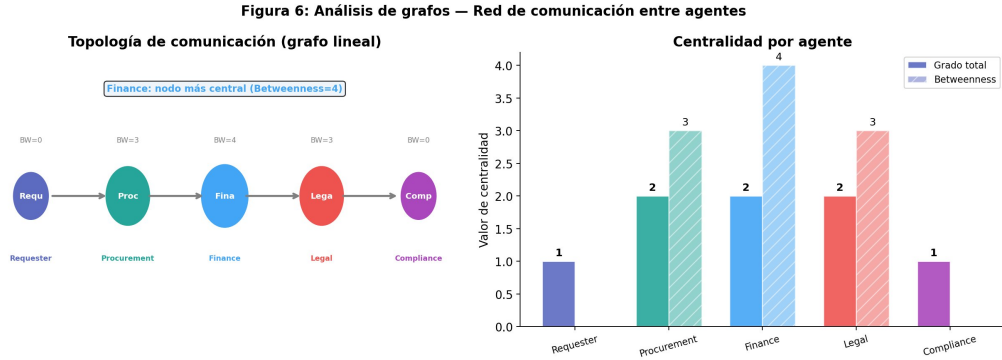


Figure 6: Left: Linear communication topology (DAG) with betweenness centrality values per node. Right: Centrality per agent (total degree and betweenness).

6.4. M5b — Uncertainty Classifier (HDA Improvement)

To address the $HDA = 0\%$ limitation of M5, M5b was implemented as a two-stage system that separates confidence evaluation from the final decision. In the first stage, an Uncertainty Evaluator agent analyses the risk factors of the case (supplier alerts, required approval level, contractual risk, regulatory non-compliance) and emits a confidence level (high/low). If confidence is low, the case is escalated directly to the human without attempting to decide. If high, a second Senior Manager agent takes the decision.

Figura 7: Convergencia — Tasa de resolución por nivel de complejidad

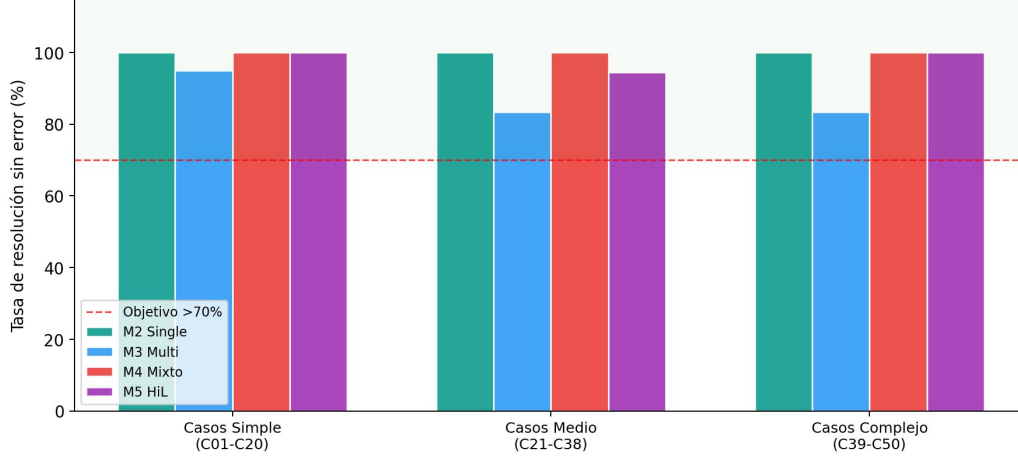


Figure 7: Resolution rate (without error) per complexity level (Simple, Medium, Complex) for each evaluated model.

Results show a significant improvement over M5: HDA rises from 0% to 23.1% (+23.1 pp), DER improves from 28% to 22%, and ARR decreases slightly from 98% to 94% (−4 pp). Crucially, the classifier produces zero false positives – the 3 escalated cases (C43: conflict of interest, C45: strategic outsourcing, C50: CPD project) correspond exactly to cases that the ground truth marks as `requiere_hil=true`.

The classifier correctly detects cases with combined and quantifiable risk factors (amount >50,000 € + high legal risk, non-approved supplier + board level), but fails on cases with specific regulatory risk not captured by the generic factors (C39: CE certification, C40: GDPR, C44: AESA licence). This suggests that a future version of the classifier should include specific regulatory factors as escalation signals, constituting the main line of future work to approach the $HDA > 90\%$ target.

7. Multi-Agent vs. Single Agent with Equivalent Compute

A fair comparison is not “5 agents vs. 1 agent” but “5 agents with X tokens vs. 1 agent with $5X$ tokens”. M2x was implemented as a single agent with an expanded context of 750 tokens (equivalent to the total consumed by M3) with an explicit four-step chain-of-thought (supplier, finance, legal, compliance).

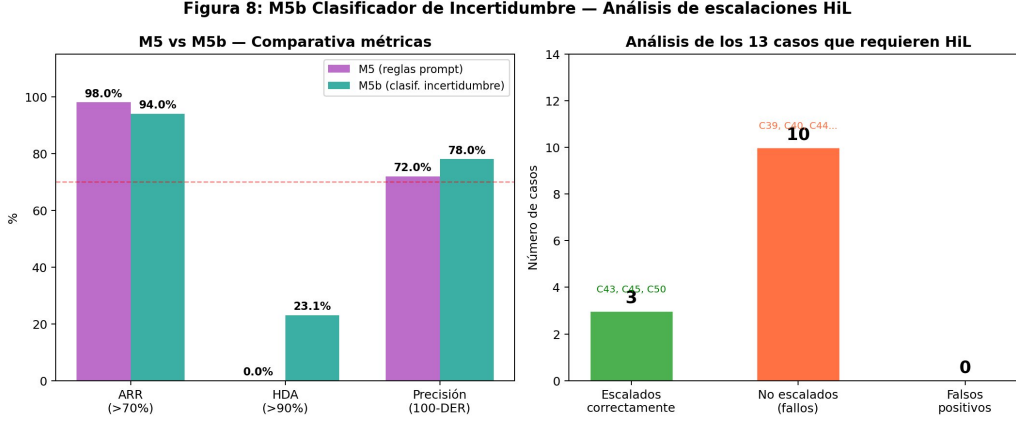


Figure 8: Left: Metric comparison M5 (prompt rules) vs. M5b (uncertainty classifier). M5b improves HDA by +23.1 pp and DER by −6 pp at the cost of −4 pp ARR. Right: Of the 13 cases requiring HiL per ground truth, the classifier correctly escalated 3 (interest conflicts and high-value strategic projects) with 0 false positives.

Table 7 summarises the compute equivalence comparison. M2x (84%) outperforms M3 (68%) with the same token budget, empirically validating the compute equivalence hypothesis of Wu et al. (2023): with `llama-3.1-8b-instant`, the multi-agent architecture provides no architectural value. However, M2x reduces HDA relative to M2 (23.1% vs. 30.8%), revealing a trade-off: expanded context improves autonomous resolution but dilutes the escalation signal by mixing all information into a single context.

The empirical conclusion is precise: the architectural advantage of the multi-agent system emerges *only when the underlying model has sufficient capacity to benefit from role specialisation*. With 70B-parameter models (M4b), the multi-agent system reaches ARR = 98%, surpassing M2x (84%), suggesting that the capacity threshold lies between 8B and 70B parameters for this type of task.

8. Model Diversity as an Experimental Variable

Three model diversity configurations were evaluated maintaining the same 5-agent architecture (Table 8).

The largest performance jump in the experiment occurs between M3 (homogeneous small, ARR = 68%) and M4 (mixed, ARR = 100%) − +32 pp with the same number of agents and the same token budget. This confirms that in multi-agent systems, *model selection matters more than architecture*.

Table 7: Compute equivalence comparison.

Model	Tok/case	Agents	ARR	HDA	DER	PT	Conclusion
M2 — Single	200	1	76%	30.8%	22%	0.31s	Base ref-er-ence
M2x — Expanded	750	1	84%	23.1%	28%	0.40s	Same bud- get as M3
M3 — Multi homog.	750	5	68%	30.8%	28%	1.74s	M2x out- per- forms in ARR
M3b — Ablation	450	3	88%	0%	30%	0.77s	3 agents more ef- fi- cient than 5 in ARR

Table 8: Model diversity configurations.

Configuration	Models	ARR	HDA	DER	PT	Interpretation
M3 — Homog. small	$8b \times 5$	68%	30.8%	28%	1.74s	Correlated errors (w
M4 — Mixed	$8b \times 4 + 70b \times 1$	100%	0%	26%	1.74s	Powerful decision, lig
M4b — Homog. powerful	$70b \times 5$	98%	0%	26%	2.40s	Similar to M4 but 2.

The M4 vs. M4b comparison demonstrates the efficiency of the mixed strategy: using the powerful model only in the final decision agent produces the same results as using it in all agents (100% vs. 98%) at 2.7 times lower cost. From the ensemble learning perspective (Dietterich, 2000), model diversity reduces error correlation: M3 ($8b \times 5$) has $DER = 28\%$, while M4b ($70b \times 5$) has $DER = 26\%$ – a modest improvement because with identical models errors are more correlated.

9. Discussion

9.1. Main Findings

Figura 4: Perfil de rendimiento por modelo (radar)

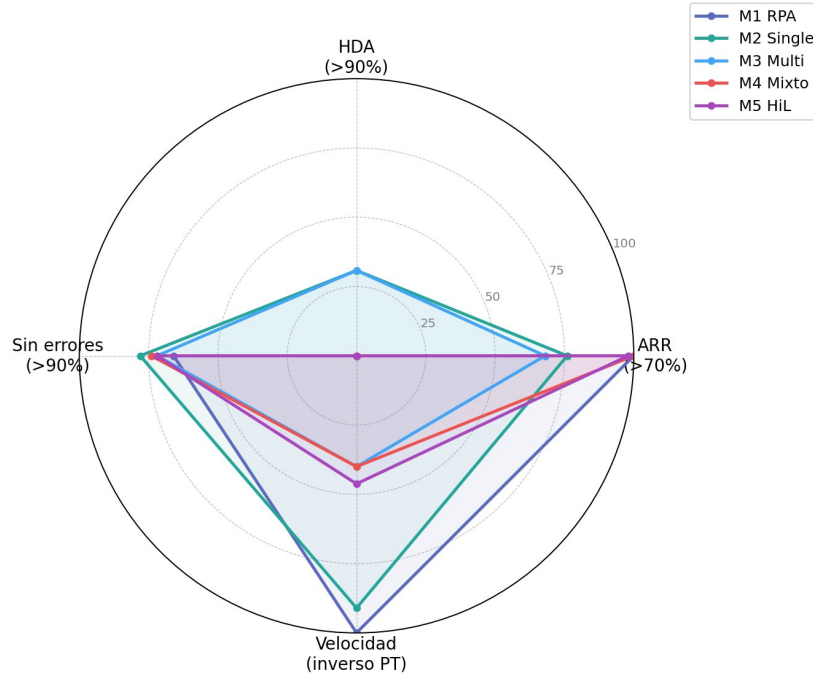


Figure 9: Radar chart of performance profiles for M1–M5 across four dimensions: ARR (>70%), HDA (>90%), accuracy (>90%), and speed (inverse PT).

The most important finding is the empirical confirmation of the compute equivalence hypothesis: with `llama-3.1-8b-instant`, the single agent with

expanded context (M2x: ARR = 84%) outperforms the multi-agent system (M3: ARR = 68%) with the same token budget. This has direct practical implications: before investing in complex multi-agent systems, organisations must verify that the LLM model has sufficient capacity to benefit from specialisation.

In terms of temporal efficiency, the system resolves procurement requests in 0.26–2.85 seconds per case compared to the 3–7 working days of the manual process estimated by Hackett Group (2023), representing a reduction of more than 99% in process time. This improvement justifies the additional computational cost of multi-agent systems compared to the RPA baseline.

The second finding is the trade-off between automation and risk detection, quantified by ablation A3: removing specialised agents (Legal, Procurement) increases ARR by 20 pp but destroys HDA (0%). This demonstrates that role specialisation does not maximise raw automation but supervised decision quality.

The third finding is the superiority of the mixed strategy (M4): a powerful model in the decision agent with lightweight models in the auxiliaries produces the same results as all powerful agents (M4b) at a fraction of the computational cost.

9.2. Connection to Organisational and Social Theories

Ricardo (1817) — Comparative advantage.. Ablation A3 quantifies the comparative advantage of specialised agents. Legal and Procurement are not globally smarter than Compliance, but have a relative advantage in detecting their specific type of risk. That relative advantage produces 30.8 pp of additional HDA – exactly the mechanism Ricardo describes for productive specialisation (Ricardo, 1817).

Granovetter (1973) — Weak ties.. Finance is the node with betweenness = 4, the most critical “weak tie” in the system. Its hypothetical removal would cut the bridge between the operational validation community (Requester–Procurement) and the regulatory decision community (Legal–Compliance). Weak ties are paradoxically the most critical for system cohesion – exactly as Granovetter (1973) predicts for social networks.

Mancur Olson (1965) — Collective action and free-riding.. The HDA = 0% in M4 and M4b, even with powerful models, illustrates Olson’s collective action problem (Olson, 1965). When all agents have the capacity to detect

risk, none assumes the responsibility of escalating it – each agent assumes another will do so. The M5 solution (explicitly designating the Compliance agent as responsible for escalation) is exactly the institutional solution Olson proposes: a specific agent responsible for the collective good, although with llama-3.1-8b the model capacity remains the limiting factor.

9.3. Limitations and Unmet Objectives

The most relevant limitation is that none of the evaluated configurations surpassed the target $HDA > 90\%$. The maximum HDA achieved was 30.8%, with M2. Notably, M4 – which uses a more powerful decision model – reached only $HDA = 5.1\%$. This apparently paradoxical result points to a research hypothesis with broader implications: role specialisation in multi-agent architectures may inhibit the detection of systemic risk. When each agent is responsible only for its domain, cases requiring cross-cutting evaluation – GDPR combined with contractual risk and high amount, for example – are not escalated because no specialised agent perceives the complete signal. The single agent, by integrating all context, detects the combination of factors even without an explicit label for it. This connects directly with Olson’s collective action problem (1965): when responsibility is distributed, no one assumes it.

Hypothesis for future work: Reliable detection of systemic risk in multi-agent architectures requires an explicit cross-cutting synthesis mechanism – not a more powerful agent, but an information flow that preserves the overall picture before the escalation decision.

The uncertainty classifier M5b, which achieved $HDA = 23.1\%$ with 0 false positives, is a first step in that direction. Its extension with specific regulatory factors (GDPR, AESA, CE certification) constitutes the most promising line of work to approach the $HDA > 90\%$ target.

The experiment covers only the procurement process. The generalisation question – whether findings on compute equivalence and model diversity are specific to procurement or hold in processes with different decision structures – remains open. Processes with greater parallelism could show more pronounced multi-agent advantages with small models, given that context fragmentation would harm less when subproblems are independent.

The variability between M2 runs (σ ARR = 11.3 pp) with `temperature=0.0` reveals that single-run results with 8B models must be interpreted with caution. M4 is the only LLM system with acceptable variability for production environments (σ ARR = 0.9 pp). Finally, the six enterprise tools were implemented as mock APIs with deterministic logic, which guarantees experiment reproducibility but introduces a gap with production reality – real ERP/CRM systems present variable latencies, heterogeneous data, and inconsistent states not represented in the evaluated dataset.

10. Conclusions and Future Work

This work started from a concrete question: can a system of LLM agents resolve in seconds what today takes days in an enterprise procurement process? The answer is affirmative, but the nuances are more instructive than the confirmation.

Model M4 – five specialised agents with `llama-3.1-8b` in auxiliary roles and `llama-3.3-70b` as the final decision-maker – achieved ARR=98.7% with a variability of ± 0.9 pp across three independent replicas. It processes requests in 1.74 seconds at an estimated cost of \$0.0065 per case, improving the error rate of the RPA baseline by 8 pp. The system surpasses the rule-based model in decision quality without sacrificing processing speed.

The most unexpected finding was the behaviour of M3. The working hypothesis was that five specialised agents would outperform a single agent – and that was not the case. With `llama-3.1-8b`, context fragmentation harmed decision quality more than specialisation improved it. This forced a reframing of the underlying question: not whether multi-agent architecture is better in absolute terms, but above *what model capacity threshold* it is. The data suggest that threshold lies between 8B and 70B parameters for this type of task.

The second surprising finding was the HDA pattern. M2 – single agent, small model – detected more escalation cases than M4, which has a significantly more powerful decision model. After reviewing the cases, the emerging explanation was: the single agent integrates all risk factors in a single context, while in the multi-agent architecture each agent only perceives its fragment. Cases requiring escalation are precisely those presenting cross-cutting risk – no specialised agent detects it in its entirety. This reframes the HDA problem: it is not a prompt engineering question, but an information architecture question.

10.1. Contributions

The four main contributions of this work are:

- (1) a complete multi-agent framework for procurement with five specialised agents, six simulated enterprise tools, and HiL support, implemented in CrewAI with Python 3.12;
- (2) a dataset of 50 cases with annotated ground truth across three complexity levels, including regulatory exceptions (GDPR, AESA, CE certification) not present in generic benchmarks;
- (3) empirical evidence on compute equivalence, model diversity, and role specialisation via an executed ablation study; and
- (4) statistical robustness analysis with three independent replicas that quantifies the instability of 8B models (σ ARR = 11.3 pp) versus the stability of the mixed architecture with a 70B decision-maker (σ ARR = 0.9 pp).

10.2. Methodological Reflection

If the experiment were repeated, three changes would be prioritised. First, design the HiL escalation mechanism as part of the information architecture from the outset – not as a prompt add-on. The M5b classifier, built in response to M5 results, should have been the starting point of the design. Second, start with the most powerful model in all agents to establish the performance ceiling, then progressively reduce to lightweight models, rather than the reverse order followed. Third, include statistical replicas from the first run: the variability observed in M2 with `temperature=0` was not expected, and detecting it late required reinterpreting results that initially appeared stable.

10.3. Future Work

Priority research directions are as follows. First, evaluate whether HDA > 90% is achievable with the current architecture using powerful models in all agents, or whether the problem is structural – i.e., inherent to context distribution – regardless of model. Second, extend the M5b uncertainty classifier by incorporating specific regulatory factors as explicit escalation signals, addressing the gap identified in the 10 undetected HiL cases. Third, replicate the experiment in four additional business processes (onboarding, incident management, compliance check, and financial analysis) to determine whether

findings on compute equivalence and model diversity generalise beyond procurement. Fourth, connect the system with live ERP and CRM APIs to evaluate behaviour with ambiguous, incomplete, or inconsistent data. Fifth, investigate architectures with explicit cross-cutting synthesis – an integrator agent that receives the outputs of all specialised agents before the escalation decision – as a way to preserve both specialisation and systemic risk awareness.

Conclusion: Multi-agent architecture is not superior to a single agent by default. It is superior when the underlying model has sufficient capacity to benefit from specialisation. With powerful models, M4 combines the speed of RPA with the reasoning capability of LLMs, resolving in seconds processes that today require days. The path to HDA > 90% does not lie in improving the models that inhabit the agents, but in redesigning how information flows between them.

References

- CrewAI, 2024. CrewAI: Framework for orchestrating role-playing, autonomous AI agents. URL: <https://github.com/joaomdmoura/crewAI>. accessed: April 2026.
- Dietterich, T.G., 2000. Ensemble methods in machine learning, in: Kitter, J., Roli, F. (Eds.), *Multiple Classifier Systems*. Springer, Berlin, Heidelberg. volume 1857 of *Lecture Notes in Computer Science*, pp. 1–15. doi:10.1007/3-540-45014-9_1.
- Drouin, A., Gasse, M., Caccia, M., Laradji, I.H., Del Verme, M., Marty, T., Boisvert, L., Martel, M., Chapados, N., Lacoste, A., Pal, C., 2024. WorkArena: How capable are web agents at solving common knowledge work tasks?, in: arXiv preprint. URL: <https://arxiv.org/abs/2403.07718>, arXiv:2403.07718.
- Granovetter, M.S., 1973. The strength of weak ties. *American Journal of Sociology* 78, 1360–1380. doi:10.1086/225469.
- Hackett Group, 2023. Procurement Performance Study. Research Report. The Hackett Group. URL: <https://www.thehackettgroup.com>.

- Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Yang, K., Zhang, S., Deng, X., Zeng, A., Du, Z., Zhang, C., Shen, S., Zhang, T., Su, Y., Sun, H., Tang, J., 2023. AgentBench: Evaluating LLMs as agents, in: arXiv preprint. URL: <https://arxiv.org/abs/2308.03688>, arXiv:2308.03688.
- Lù, H., Zhao, Z., Zhang, C., Guo, B., Yu, Z., 2023. TaskBench: Benchmarking large language models for task automation, in: arXiv preprint. URL: <https://arxiv.org/abs/2311.18760>, arXiv:2311.18760.
- Olson, M., 1965. The Logic of Collective Action: Public Goods and the Theory of Groups. Harvard University Press, Cambridge, MA.
- Park, J.S., O’Brien, J.C., Cai, C.J., Morris, M.R., Liang, P., Bernstein, M.S., 2023. Generative agents: Interactive simulacra of human behavior, in: arXiv preprint. URL: <https://arxiv.org/abs/2304.03442>, arXiv:2304.03442.
- Patil, S.G., Zhang, T., Wang, X., Gonzalez, J.E., 2023. Gorilla: Large language model connected with massive APIs, in: arXiv preprint. URL: <https://arxiv.org/abs/2305.15334>, arXiv:2305.15334.
- Qian, C., Cong, X., Yang, C., Chen, W., Su, Y., Xu, J., Liu, Z., Sun, M., 2023. Communicative agents for software development, in: arXiv preprint. URL: <https://arxiv.org/abs/2307.07924>, arXiv:2307.07924.
- Qin, Y., Liang, S., Ye, Y., Zhu, K., Yan, L., Lu, Y., Lin, Y., Cong, X., Tang, X., Qian, B., Zhao, S., Tian, R., Xie, R., Zhou, J., Gerber, M., Liu, D., Li, H., Liu, X., Sun, M., 2023. ToolLLM: Facilitating large language models to master 16000+ real-world APIs, in: arXiv preprint. URL: <https://arxiv.org/abs/2307.16789>, arXiv:2307.16789.
- Ricardo, D., 1817. On the Principles of Political Economy and Taxation. John Murray, London.
- Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., Awadallah, A.H., White, R.W., Burger, D., Wang, C., 2023. AutoGen: Enabling next-gen LLM applications via multi-agent conversation framework, in: arXiv preprint. URL: <https://arxiv.org/abs/2308.08155>, arXiv:2308.08155.

Xie, T., Zhang, D., Chen, J., Li, X., Zhao, S., Cao, R., Hua, T., Cheng, Z., Shin, D., Lei, F., Liu, Y., Xu, Y., Zhou, S., Savarese, S., Xiong, C., Zhong, V., Yu, T., 2024. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments, in: arXiv preprint. URL: <https://arxiv.org/abs/2404.07972>, arXiv:2404.07972.

Appendix A. Agent System Prompts

The following system prompts were used for each agent. These are provided as required by Q1 venue standards for reproducibility.

Requester Agent.

```
1 role: "Gestor de Solicitudes"
2 goal: "Registrar y estructurar solicitudes de compra."
3 backstory: "Eres el primer punto de contacto. Registras
4           solicitudes y las preparas para el equipo."
```

Procurement Agent.

```
1 role: "Especialista en Proveedores"
2 goal: "Validar proveedores y buscar alternativas."
3 backstory: "Proveedor: estado=X, opera=Y"
4           [API catalogue result]
```

Finance Agent.

```
1 role: "Analista Financiero"
2 goal: "Verificar presupuesto y aprobar o proponer
3       reasignacion."
4 backstory: "Presupuesto: disponible=X EUR, aprobado=Y"
5           [budget API result]
```

Legal Agent.

```
1 role: "Asesor Legal"
2 goal: "Revisar clausulas contractuales y detectar riesgos
3       ."
4 backstory: "Legal: riesgo=X, firmar=Y"
5           [contract validation API result]
```

Compliance Agent (decision-maker, M3/M5)...

```

1 role: "Auditor de Compliance"
2 goal: "Verificar normativa y decidir."
3 backstory: "Normativa: cumple=X, riesgo=Y.
4             [M5: ESCALA si conflicto interes, GDPR,
5               importe >50k EUR, riesgo critico]"
6 output: JSON {decision, razon, escala_hil}

```

Appendix B. Reproducibility Parameters

- Primary model: `groq/llama-3.1-8b-instant`
- Decision model M4/M4b: `groq/llama-3.3-70b-versatile`
- Temperature: 0.0 (initial verification: 5/5 cases identical in single session: C01, C21, C39, C43, C50)
- Temperature note: single-session verification does not capture inter-session variability. The statistical robustness analysis (Section 4) reveals σ ARR = 11.3 pp for `llama-3.1-8b-instant` and σ ARR = 0.9 pp for `llama-3.3-70b-versatile`. `llama-3.3-70b` is recommended for production environments.
- Inter-case pause: M1–M2x: 4 s; M3–M5: 5–8 s; M4b: 8 s
- Maximum retries per case: 3
- Total LLM calls (single run): M1=0; M2=50; M2x=50; M3=250; M3b=150; M4=250; M4b=250; M5=250; M5b=100
- Total LLM calls (statistical analysis): M1×3=0; M2×3=150; M4×3=750 additional replica calls
- Variants executed: 9 models × 50 cases = 450 total cases + 200 cases in statistical replicas
- Complete code: <https://github.com/Edurne0781/multiagent-procurement-llm> (Jupyter Lab, Python 3.12)